

An-Najah National University

Faculty of Engineering and Information Technology

Data Warehousing Assignment

Movie Rental Data Warehouse Design

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1 Introduction

This assignment presents a high-level data warehouse design for a movie rental business based on the Sakila OLTP database.

The OLTP system is designed for daily operational transactions such as rentals, payments, customer registration, and inventory management. Although it is efficient for transaction processing, it is not suitable for analytical reporting because the data is highly normalized and distributed across many tables.

The purpose of the proposed data warehouse is to reorganize operational data into a dimensional model that supports business analysis and decision-making.

The dimensional model simplifies querying and allows managers to analyze revenue, rentals, customer activity, film popularity, and store performance more efficiently.

The design uses a star schema approach with fact tables and shared dimension tables.

2 Business Questions

The proposed data warehouse is designed to answer important analytical business questions related to the movie rental business.

1. Which films are rented most frequently?
2. Which films generate the highest revenue?
3. Which stores generate the highest number of rentals?
4. Which customers generate the highest revenue?
5. How does rental activity change over time?
6. Which staff members process the highest number of rentals or payments?
7. What is the average rental duration for films?
8. Which films are returned late most often?

These questions are important because they help managers understand customer behavior, identify profitable films, evaluate store performance, and monitor rental trends over time.

The analysis also supports better business decisions related to marketing, inventory management, customer service, and operational efficiency.

3 Dimensional Model Design

3.1 Selected Business Processes

The main business processes selected from the OLTP system are:

- Rental transactions
- Payment transactions
- Film performance analysis

These processes represent the main operational activities that require analytical reporting.

The proposed dimensional model uses a star schema because it simplifies reporting queries and improves analytical performance.

3.2 Fact Tables

3.2.1 Fact_Rental

Business Process:

Rental transaction analysis.

Grain:

One row for each rental transaction.

Measures:

- rental_count
- rental_duration
- late_return_days

Related Dimensions:

- Dim_Date
- Dim_Customer
- Dim_Film
- Dim_Store
- Dim_Staff

3.2.2 Fact_Payment

Business Process:

Payment and revenue analysis.

Grain:

One row for each payment transaction.

Measures:

- payment_amount
- payment_count

Related Dimensions:

- Dim_Date
- Dim_Customer
- Dim_Film
- Dim_Store
- Dim_Staff

3.2.3 Fact_Film_Performance

Business Process:

Film performance and rental trend analysis.

Grain:

One row for each film per store per day.

Measures:

- total_rentals
- total_revenue
- average_rental_duration

Related Dimensions:

- Dim_Date
- Dim_Film
- Dim_Store

3.3 Dimension Tables

3.3.1 Dim_Date

Main Attributes:

- date_key
- full_date
- day_number
- month_number
- month_name
- quarter_number
- year_number

Source OLTP Tables:

Rental and payment date fields.

Purpose:

Used for time-based analysis and trend reporting.

3.3.2 Dim_Customer

Main Attributes:

- customer_key
- customer_id
- full_name
- email
- active_status
- city
- country

Source OLTP Tables:

customer, address, city, country

Purpose:

Used to analyze customer activity and revenue generation.

3.3.3 Dim_Film

Main Attributes:

- film_key
- film_id
- title
- category_name
- language_name
- rental_rate
- rental_duration
- release_year

Source OLTP Tables:

film, category, film_category, language

Purpose:

Used to analyze film popularity and revenue.

3.3.4 Dim_Store

Main Attributes:

- store_key
- store_id
- city
- country
- manager_name

Source OLTP Tables:

store, address, city, country, staff

Purpose:

Used to analyze store performance and location activity.

3.3.5 Dim_Staff

Main Attributes:

- staff_key
- staff_id
- full_name
- store_id

Source OLTP Tables:

staff

Purpose:

Used to analyze staff activity and operational performance.

4 Dimensional Model Diagram

The following diagram shows the proposed star schema for the movie rental data warehouse.

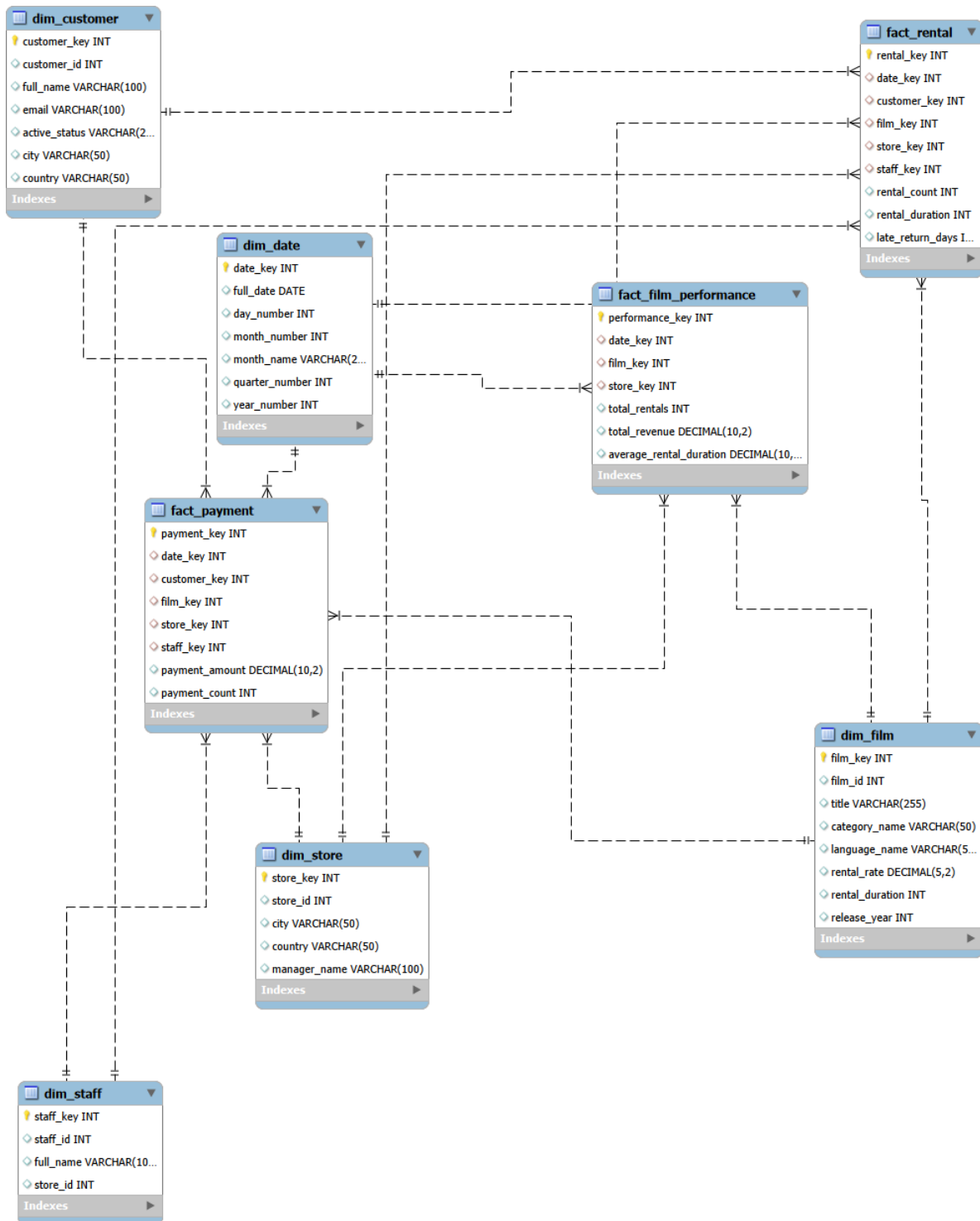


Figure 1: Star Schema for the Movie Rental Data Warehouse

The dimensional model contains shared dimensions between multiple fact tables. Shared dimensions include:

- Dim.Date
- Dim.Customer

- Dim.Film
- Dim.Store
- Dim.Staff

The main fact tables are:

- Fact.Rental
- Fact.Payment
- Fact.Film_Performance

The star schema simplifies analytical queries and improves reporting performance.

5 ETL Design

5.1 Extract

The extraction process collects data from the OLTP system tables required for the dimensional model.

The main extracted tables include:

- rental
- payment
- customer
- film
- inventory
- store
- staff
- address
- city
- country
- category
- film_category
- language

These tables provide the operational data needed to build dimensions and fact tables.

5.2 Transform

Several transformations are applied before loading the warehouse.

- Related OLTP tables are joined to create complete dimensions.
- First name and last name are combined into full names.
- Missing or duplicate records are cleaned.
- Surrogate keys are generated for dimension tables.
- Date keys are created from rental and payment dates.

- Rental duration and late return days are calculated.
- Payment amounts are aggregated for revenue analysis.
- Text values such as country and city names are standardized.
- Null return dates are handled for films not returned yet.

These transformations prepare the data for analytical reporting.

5.3 Load

The loading process inserts transformed data into the data warehouse.

Dimension tables are loaded first because fact tables depend on their surrogate keys.

The recommended loading order is:

1. Dim_Date
2. Dim_Customer
3. Dim_Film
4. Dim_Store
5. Dim_Staff
6. Fact_Rental
7. Fact_Payment
8. Fact_Film_Performance

Fact tables reference dimension surrogate keys to maintain relationships between facts and dimensions.

Data validation checks are applied during loading to ensure consistency and accuracy.

6 Data Quality Considerations

To ensure reliable reporting and accurate analysis, several data quality checks should be applied during the ETL process.

- Missing values should be detected and handled before loading the warehouse.
- Duplicate customer, payment, or rental records should be removed.
- Rental return dates must not be earlier than rental dates.
- Payment amounts should always be positive values.
- Customer names, city names, and country names should be standardized.
- Invalid or incomplete records should be rejected during loading.
- Foreign key relationships between fact tables and dimensions should be validated.
- Null return dates should be handled properly for rentals that were not returned yet.

These quality checks improve data consistency and help managers make accurate business decisions using the data warehouse.

7 Analytical Reports and Visualizations

The following charts present analytical insights generated from the data warehouse after executing ETL processes and analytical SQL queries.

These visualizations help management understand rental behavior, store performance, and customer activity trends.

7.1 Top Countries by Rentals

The following chart shows the countries with the highest number of movie rentals.

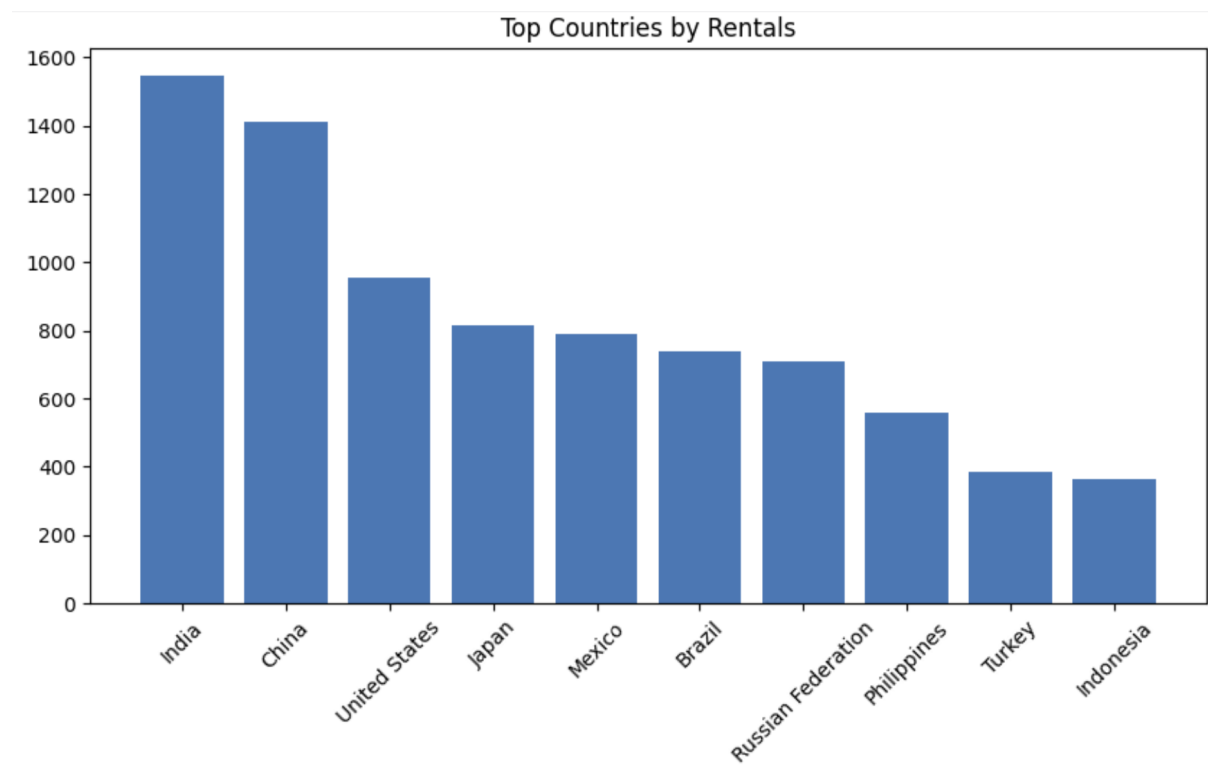


Figure 2: Top Countries by Rentals

This analysis helps identify the most active customer regions and supports location-based business analysis.

7.2 Rentals per Store

The following chart compares the total rental activity between stores.

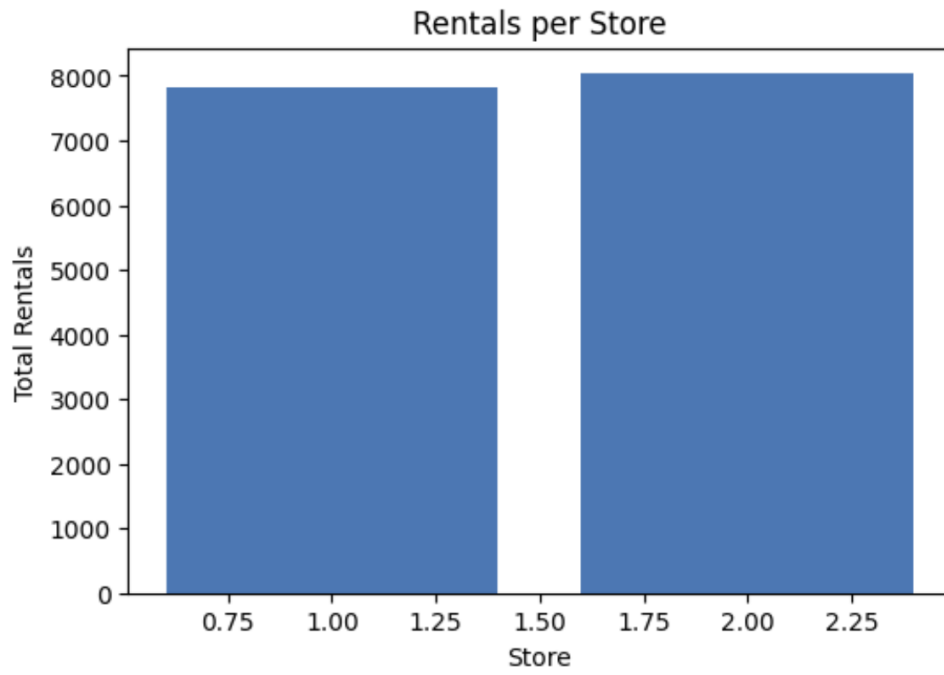


Figure 3: Rentals per Store

The chart helps evaluate store performance and compare rental demand across branches.

7.3 Top Rented Movies

The following chart presents the most rented movies in the system.

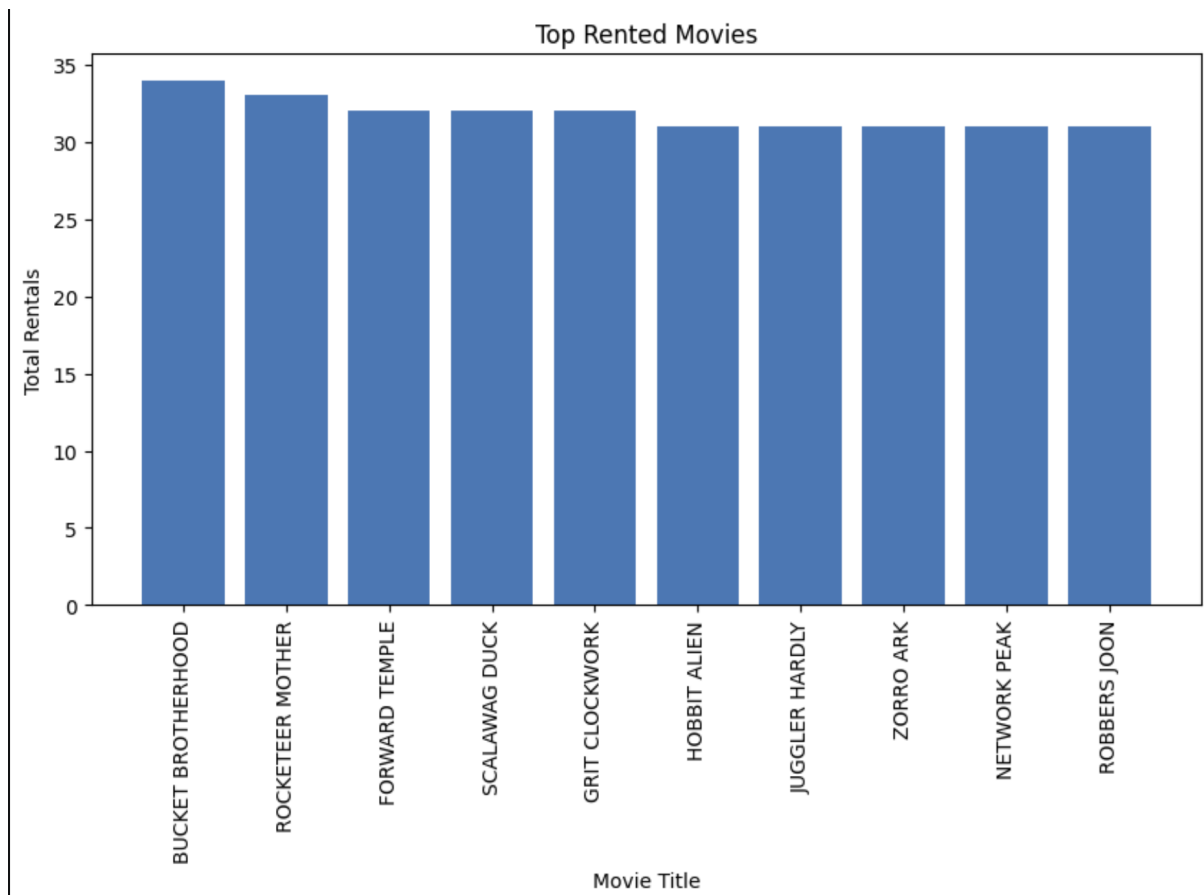


Figure 4: Top Rented Movies

This analysis helps management identify popular films and improve inventory and marketing decisions.

8 Conclusion

This assignment presented a high-level data warehouse design for the movie rental system based on the Sakila OLTP database.

The proposed dimensional model reorganizes operational data into fact tables and dimension tables that support analytical reporting and business intelligence.

The designed data warehouse allows managers to analyze rentals, revenue, customer behavior, film popularity, store performance, and time-based trends more efficiently.

The ETL process helps extract data from the OLTP system, clean and transform the data, and load it into the warehouse in a structured format.

Using a star schema design improves query performance, simplifies reporting, and supports better business decision-making.

Overall, the proposed solution provides a clear and scalable foundation for analytical processing in the movie rental business.